

ConstructionSafety-FaithBench: Counterfactual Evaluation of Visual-Evidence Faithfulness in Vision-Language Safety Inspection

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Abstract

Construction-safety vision-language models are commonly evaluated by final answers, textual rationales, or grounding overlap. These signals can overstate whether a system uses rule-relevant visual evidence. We introduce the working scaffold for ConstructionSafety-FaithBench, a benchmark and audit protocol for separating answer correctness from visual-evidence faithfulness under targeted and matched-control interventions. The current repository implements the pilot benchmark manifest, model-assisted bootstrap annotations, normalized model-output schema, deterministic baselines, intervention specifications, and scoring code. The current annotation layer is not human ground truth; it is a source-metadata-assisted bootstrap used to validate the pipeline before independent domain annotation and multi-model VLM experiments.

1 Introduction

Vision-language models may support construction-safety review by linking site imagery to natural-language safety rules. However, a correct or stable final answer does not establish that the model relied on the right visual evidence. This gap matters in safety inspection: a warning is less useful if the cited evidence moves from a hard hat to unrelated background scaffolding, or if the model ignores a counterfactual edit to the rule-relevant object.

The current pilot study found that targeted occlusion changed answers more often than same-size random controls and that IoU-based drift can be distorted by object size. These observations motivate a broader benchmark contribution rather than a single-model feasibility report.

ConstructionSafety-FaithBench is designed to evaluate whether model answers, evidence regions, and rationales remain aligned under counterfactual interventions. The benchmark is currently implemented as a reproducible pilot scaffold; the next research step is to expand it to multiple VLMs, larger splits, and independent human/domain annotation.

Current contributions in this repository. The repository currently provides: (i) a normalized construction-safety rule specification, (ii) a 163-example pilot manifest, (iii) model-assisted bootstrap annotations, (iv) normalized model-output schema and deterministic baselines, (v) frozen targeted and matched-random intervention specs, and (vi) scoring scripts for answer and evidence metrics.

2 Related Work

Construction-safety VLMs. Recent construction-safety VLM work evaluates hazard identification, safety-rule VQA, grounding, and caption quality. ConstructionSite 10k is the closest dataset foundation for this project because it includes construction images, rule-violation records, captions,

and visual grounding metadata [1]. Florence-2 is relevant to the current pilot because it supports prompted grounding and detection-style outputs [4].

Faithfulness and rationales. Prior XAI and VLM literature shows that plausible explanations are not necessarily faithful. Attention, textual rationales, and final-answer correctness must therefore be treated as separate evidence channels [2]. Counterfactual and perturbation-based evaluation is especially relevant when the claim concerns whether a model uses the cited visual evidence.

Evaluation and datasets. The intended top-tier framing is an evaluation artifact: a benchmark, intervention protocol, scoring suite, and failure taxonomy. This aligns more closely with an Evaluations & Datasets contribution than with a new-model contribution unless a faithfulness-improving method is added later. The safety framing follows the broader risk-management emphasis on documented limits, measurement, and appropriate use [3].

3 Benchmark

Each benchmark example consists of an image identifier, source dataset revision, safety rule, expected answer seed, evidence metadata, and annotation fields. The first implemented rule families are hard-hat/PPE compliance, harness/fall protection, guardrail/edge protection, and struck-by/equipment proximity.

The pilot manifest contains 163 image-rule pairs from the ConstructionSite test split. It preserves both legacy rule IDs and normalized benchmark rule IDs so the benchmark remains traceable to the original pilot artifacts.

The scale-up candidate manifest contains 588 test-split image-rule pairs selected from source metadata without redistributing raw images. Table 1 shows the current class counts. The split remains imbalanced because struck-by and fall-hazard cases are scarce under the priority-labeling policy.

Table 1: Scale-up candidate manifest class counts.

Class	Count
Compliant	250
Fall Hazard	75
Ppe Violation	250
Struck By Risk	13
Total	588

Annotation status. The current completed annotation files are model-assisted bootstrap annotations generated from ConstructionSite metadata, captions, source rule-violation records, and pilot target boxes. They are useful for pipeline validation and triage, but they are not independent human ground truth. A NeurIPS-level version must add domain-audited annotations with agreement statistics.

4 Metrics

The benchmark separates answer-level metrics from evidence-level metrics.

Answer metrics. The current scoring code reports answer accuracy, macro-F1 over compliant and violation labels, invalid-output rate, and nonambiguous-example accuracy.

Evidence metrics. The current scoring code reports evidence presence and best evidence IoU against the bootstrap reference boxes. The broader intervention audit also tracks normalized centroid drift and disappearance, which are needed because IoU is size-sensitive for small PPE objects.

Intervention metrics. Each row with a target box receives one targeted occlusion and five same-size matched-random occlusions. The key comparison is whether the rule-relevant intervention changes answers or evidence more than the matched control.

5 Experimental Setup

The current Phase 3 harness provides deterministic baselines to validate the benchmark pipeline before expensive VLM experiments:

- **Annotation bootstrap:** echoes the model-assisted annotation labels and evidence.
- **Florence grounding:** runs the local Florence-2 grounding pipeline from the companion XAI pilot and maps grounded detections through a deterministic geometric decision layer.
- **Manifest seed:** echoes the pilot weak label from the manifest.
- **Majority violation:** image-blind baseline that always predicts violation.
- **Caption keyword:** caption-only leakage baseline using rule-specific keywords and no image pixels.

Future experiments should add instruction-following VLM adapters and, if budget permits, closed VLM adapters behind the same normalized output schema. Each adapter must record model version, prompt ID, decoding settings, access date, and raw response provenance.

6 Current Pilot Results

Table 2 reports the current pilot scores against the model-assisted bootstrap annotation layer. These are pipeline-validation numbers, not final benchmark claims.

Table 2: Pilot baseline scores against model-assisted bootstrap annotations.

Adapter	Accuracy	Macro-F1	Evidence present	Nonambig. acc.
Annotation bootstrap	100.0%	100.0%	98.8%	100.0%
Florence grounding	41.7%	41.7%	96.9%	37.7%
Manifest seed	100.0%	100.0%	0.0%	100.0%
Majority violation	69.3%	40.9%	0.0%	72.5%

Table 3 summarizes the frozen pilot intervention specifications. These specs can be consumed by future model runners to produce targeted-versus-control faithfulness results.

Table 3: Frozen pilot intervention specifications.

Intervention type	Count
Targeted Occlusion	158
Matched Random Occlusion	790
Total	948

Table 4: Pilot Florence intervention effects.

Metric	Targeted	Matched random	Paired diff.
Answer flip rate	39.2%	9.2%	30.0%
Centroid drift	0.188	0.029	0.160
Evidence disappearance	5.1%	3.7%	–

Table 4 reports the first completed intervention run for the Florence grounding adapter. Targeted rule-evidence masks produce more answer flips and larger evidence movement than same-size random-location masks.

The current repository therefore validates the benchmark plumbing: manifests, annotations, outputs, interventions, and score tables can be regenerated and checked automatically.

Table 5 reports the same adapters on the scale-up candidate manifest. The Florence row is the first image-grounded local model output in the repository. The caption-keyword baseline is intentionally included as a leakage diagnostic: high performance would suggest that source captions encode too much of the label, while low performance suggests the split cannot be solved by this simple caption-only heuristic.

Table 5: Scale-up baseline scores against model-assisted bootstrap annotations.

Adapter	Accuracy	Macro-F1	Evidence present	Nonambig. acc.
Annotation bootstrap	100.0%	100.0%	57.3%	100.0%
Florence grounding	39.3%	38.6%	98.8%	28.3%
Manifest seed	100.0%	100.0%	0.0%	100.0%
Majority violation	57.5%	36.5%	0.0%	82.7%
Caption keyword	19.7%	26.6%	0.0%	8.1%

Table 6 breaks the scale-up Florence result down by safety rule. The current local grounding pipeline is weakest on PPE and struck-by examples, showing where adapter and annotation work should be prioritized next.

7 Limitations and Ethics

The current scaffold has three major limitations. First, the completed annotation layer is model-assisted and metadata-derived, not independent human ground truth. Second, the current model outputs are deterministic baselines, not real multi-model VLM evaluations. Third, the pilot sample is only 163 image-rule pairs and should not be treated as a full benchmark.

Construction-site imagery also raises privacy, surveillance, and labor-governance concerns. This benchmark must not be used as an autonomous inspection, worker discipline, or surveillance system.

Table 6: Scale-up Florence rule slices.

Rule	N	Accuracy	Evidence present	Mean IoU
Fall harness	69	59.4%	98.6%	0.130
Guardrail edge	110	63.6%	96.4%	0.171
PPE hard hat	303	30.7%	99.3%	0.030
Struck-by equipment	106	25.5%	100.0%	0.033

Future releases should preserve source licensing, avoid redistributing images when rights are unclear, document non-commercial restrictions, and include prohibited-use language.

References

- [1] X. Chen and Z. Zou. Are large pre-trained vision language models effective construction safety inspectors?, 2025.
- [2] Sarthak Jain and Byron C. Wallace. Attention is not explanation. *Proceedings of NAACL-HLT*, 2019.
- [3] National Institute of Standards and Technology. Artificial intelligence risk management framework (ai rmf 1.0), 2023.
- [4] Bin Xiao, Haiping Wu, Weijian Xu, Xiyang Dai, Houdong Hu, Yumao Lu, Michael Zeng, Ce Liu, and Lu Yuan. Florence-2: Advancing a unified representation for a variety of vision tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

A Artifact Map

Key artifacts:

- `benchmark/rules.json`: normalized safety rules.
- `benchmark/splits/`: pilot image-rule manifest.
- `benchmark/splits/scaleup_candidate_manifest.csv`: larger candidate manifest.
- `benchmark/annotations/`: bootstrap annotation templates and completed model-assisted annotations.
- `benchmark/interventions/`: frozen intervention specs.
- `results/frozen_model_outputs/`: normalized baseline outputs.
- `results/tables/`: generated score tables.

B Regeneration Commands

```
python .\experiments\build_pilot_manifest.py
python .\experiments\build_model_assisted_annotations.py
python .\experiments\run_model_harness.py --adapter annotation_bootstrap
python .\experiments\build_intervention_specs.py
python .\experiments\score_model_outputs.py
python .\experiments\validate_phase4_outputs.py
python -m pytest .\analysis\tests .\tests -q
```